Vusumuzi Nkosi

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SUMMARY

Data science postgraduate student (STADIO, NQF 8) and **ALX Data Science graduate** who frames every analysis around the business decision it supports. Six end-to-end projects across forecasting, fraud detection, churn, A/B testing and segmentation, including a **day-ahead forecast of South Africa's electricity demand within 0.1 points of Eskom's own** and a fraud review queue that catches **98% of fraud cases** at a team's daily capacity. Nine years of mathematics teaching developed strong statistical thinking and the ability to explain results clearly to non-technical stakeholders. Seeking data scientist or data analyst roles in South Africa or remote.

TECHNICAL SKILLS

Programming: Python, SQL, R (basic)

ML & Modelling: scikit-learn, XGBoost, gradient boosting, random forest, logistic regression, K-means clustering, feature engineering, cost-based thresholds, time-based and walk-forward validation, imbalanced data

Statistics & Experimentation: hypothesis testing, A/B testing, McNemar's test, confidence intervals, power analysis, regression analysis, time-series forecasting, SciPy, statsmodels

Data & Visualisation: pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, Excel

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, AWS SageMaker, Prompt Engineering, Generative AI

PROJECTS

Account Activation Prediction — STADIO Capstone (CAP182) Python, XGBoost, scikit-learn
github.com/vusumuzinkosi/cap182-capstone

- Predicted, within 24 hours of registration, which new investing-platform accounts would not make a first deposit within 30 days, using only data available at scoring time to prevent target leakage.
- Validated on the UCI Bank Marketing dataset (41,188 rows), where XGBoost beat logistic regression (PR-AUC 0.49 vs 0.46; McNemar's $p < 0.001$), then applied to the client's 8-table extract with a time-based split.
- On unseen data (ROC-AUC 0.73), contacting the riskiest 20% gave a 1.79× lift over random outreach (95% CI 1.48–2.12) and captured 35% of stallers; presented in an 11-minute recorded video.

Eskom Demand Forecasting Python, scikit-learn, gradient boosting
github.com/vusumuzinkosi/eskom-demand-forecasting

- Forecast South Africa's hourly electricity demand a week ahead from 4.5 years of Eskom data (39,600 hours), using calendar, public-holiday and lagged-demand features with 52 weeks of walk-forward validation.
- Reached 3.1% MAPE a week ahead, 15% better than a “same as last week” baseline, and 2.4% a day ahead against Eskom's published 2.3%, without weather data.

Fraud Detection: A Daily Review Queue for Mobile Money Python, scikit-learn
github.com/vusumuzinkosi/fraud-detection-mobile-money

- Built a daily review queue for a six-analyst team from 2.6 million mobile money transactions, with a time-based split and only information known at transaction time; found and removed a leaking balance column.
- Reviewing the top 300 transactions a day catches 98% of fraud cases (89% of the money) with 87% of flags genuine, where the existing rule caught 1 fraud in 4 days.

Churn Prediction: Revenue at Risk Python, scikit-learn
github.com/vusumuzinkosi/churn-prediction-revenue-at-risk

- Trained logistic regression and random forest models with a decision threshold set from the business costs of each error; the random forest catches 95% of churners on unseen customers (ROC AUC 0.98).
- Ranking customers by expected loss protects 82% of the value at risk at 10% contact capacity, against 58% when ranking by churn probability; exported a ranked call list with a spending ceiling per customer.

A/B Test: New Landing Page Python, statsmodels, SciPy

github.com/vusumuzinkosi/ab-test-landing-page

- Pre-registered a two-sided test (α = 0.05, +0.5-point minimum worthwhile lift) on 290,584 visitors, after cleaning mismatched rows and duplicate users.
- Recommended not rolling out: the new page converted 0.16 points lower (p = 0.19, 95% CI −0.39 to +0.08), with 98% power for the lift that matters.

Customer Segmentation: RFM and K-means Python, pandas, scikit-learn

github.com/vusumuzinkosi/customer-segmentation-rfm

- Segmented 5,852 customers from two years of retail transactions with hand-built RFM tiers, then K-means on log-scaled RFM with k chosen by silhouette score and business use.
- Found loyal big spenders are 19% of customers but 72% of revenue; segments built on older data predicted the next six months, with the top group spending 19× as much as lapsed buyers.

WORK EXPERIENCE

Senior Lecturer — Mathematics & IT Feb 2017 – Present

Amajuba TVET College — Newcastle, KwaZulu-Natal

- Deliver mathematics and IT instruction to 100+ students annually across multiple cohorts, designing curriculum and assessments aligned with national standards.
- Analyse student performance data to identify at-risk learners and adjust teaching strategies, improving pass rates through targeted, data-driven interventions.
- Developed standardised assessment rubrics adopted across the mathematics department for consistent grading.
- Mentored students in IT competitions at regional and national level, with multiple teams advancing to national rounds.

Special Educator — Mathematics Jan 2012 – Mar 2016

Mdumiseni High School — KwaZulu-Natal

- Delivered mathematics lessons for diverse learner groups, tracking individual progress and adapting materials to different learning needs.

EDUCATION

Postgraduate Diploma in Data Science (NQF 8) 2026 (in progress)

STADIO Higher Education — Remote

- 120-credit qualification (SAQA ID: 122721); modules: Introduction to Data Science, Working with Data, Statistical Modelling, Applied Machine Learning, Capstone Project.

Data Science Programme Mar 2025 – Apr 2026

ALX Africa — Remote

Postgraduate Diploma in TVET 2024 – 2025

University of the Western Cape

BSc Mathematics & Computer Science 2019 – 2024

University of South Africa

BEd Mathematics & Computer Science 2012 – 2017

University of South Africa

CERTIFICATIONS

- Data Science Certificate — ALX Africa (2025–2026)
- Data Analyst Certificate — ALX Africa (2025): SQL, Power BI, A/B Testing, Regression Analysis
- AI Engineering Bootcamp — Zero To Mastery Academy (2025)
- Prompt Engineering Bootcamp — Zero To Mastery Academy (2025)